

Regression Model Selection - Ksi

Dependent variable: Ksi

Independent variables:

A=VWC

B=Soil_T

C=zrážky[mm]

D=Najvyššia teplota

E=Max. rýchlosť vetra[m/s]

F=Najnižšia vlhkosť[%]

Number of complete cases: 8

Number of models fit: 57

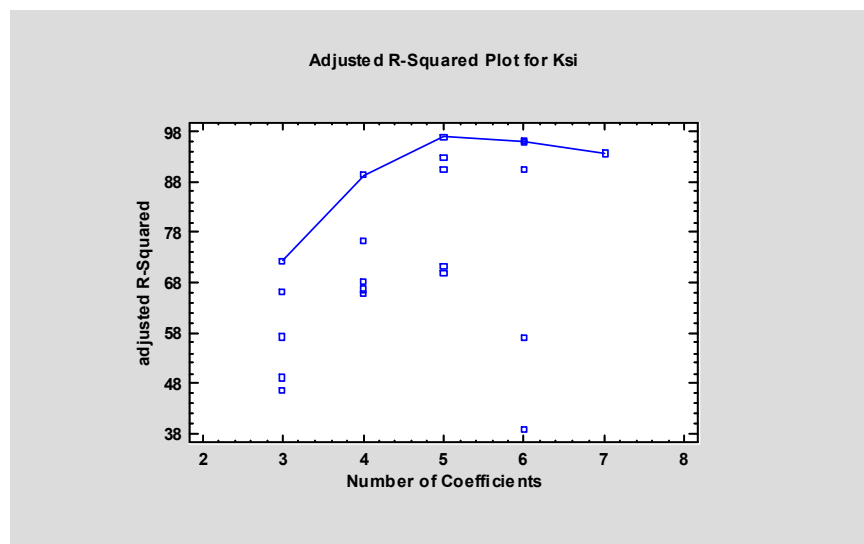
Model Results

		Adjusted		Included
MSE	R-Squared	R-Squared	Cp	Variables
0,00213819	30,3199	2,44791	73,9123	AB
0,000605707	80,261	72,3655	19,5044	AC
0,00219185	28,5714	0,0	90,8841	AD
0,00219185	28,5714	0,0	76,88	AE
0,00219185	28,5714	0,0	75,8706	AF
0,000936865	69,4692	57,2569	31,2615	BC
0,00219185	28,5714	0,0	85,458	BD
0,00219185	28,5714	0,0	79,2684	BE
0,00219185	28,5714	0,0	79,0735	BF
0,00116625	61,9938	46,7913	39,4055	CD
0,00074102	75,8514	66,192	24,3085	CE
0,00111333	63,7184	49,2057	37,5267	CF
0,00219185	28,5714	0,0	93,2509	DE
0,00219185	28,5714	0,0	88,3628	DF
0,00219185	28,5714	0,0	87,9113	EF
0,000233562	93,9109	89,3441	6,63373	ABC
0,00219185	42,8571	0,0	73,8426	ABD
0,00181727	52,6226	17,0895	51,6149	ABE
0,00219185	42,8571	0,0	63,1458	ABF
0,00074911	80,4702	65,8229	21,2765	ACD
0,00051599	86,5478	76,4587	14,6554	ACE
0,000696133	81,8514	68,2399	19,7719	ACF
0,00219185	42,8571	0,0	78,8367	ADE
0,00219185	42,8571	0,0	75,8665	ADF
0,00219185	42,8571	0,0	76,8835	AEF
0,00116879	69,529	46,6758	33,1963	BCD
0,000732686	80,8984	66,5722	20,8101	BCE
0,00100975	73,6751	53,9314	28,6795	BCF
0,00219185	42,8571	0,0	80,93	BDE
0,00219185	42,8571	0,0	80,9026	BDF
0,00219185	42,8571	0,0	80,5497	BEF
0,000802744	79,072	63,3759	22,7999	CDE
0,00125165	67,3688	42,8954	35,5498	CDF
0,000754583	80,3276	65,5732	21,432	CEF
0,00219185	42,8571	0,0	89,4243	DEF
0,000212526	95,8445	90,3038	6,52719	ABCD
0,000157573	96,919	92,811	5,35658	ABCE
0,000065393	98,7214	97,0165	3,39299	ABCF
0,00219185	57,1429	0,0	50,372	ABDE
0,00219185	57,1429	0,0	61,6868	ABDF
0,00210936	58,7558	3,76343	46,9332	ABEF
0,000658711	87,1202	69,9472	16,0318	ACDE
0,000900026	82,4018	58,9376	21,1722	ACDF
0,000628593	87,7091	71,3213	15,3902	ACEF
0,00219185	57,1429	0,0	77,6391	ADEF
0,000899079	82,4203	58,9808	21,152	BCDE
0,00118164	76,8955	46,0894	27,171	BCDF
0,000911991	82,1679	58,3917	21,4271	BCEF
0,00219185	57,1429	0,0	82,5465	BDEF
0,00100047	80,4377	54,3547	23,3119	CDEF

0,000213148	97,2216	90,2754	7,02696	ABCDE
0,0000896961	98,8308	95,9077	5,27379	ABCDF
0,0000878581	98,8547	95,9916	5,24769	ABCEF
0,00219185	71,4286	0,0	48,9209	ABDEF
0,000942875	87,7093	56,9827	17,39	ACDEF
0,0013403	82,5288	38,8507	23,0339	BCDEF
0,000140833	99,0821	93,5747	7,0	ABCDEF

The StatAdvisor

This table shows the results of fitting various multiple regression models to describe the relationship between Ksi and 6 predictor variables. Models have been fit containing all combinations of from 2 to 6 variables. The statistics tabulated include the mean squared error (MSE), the adjusted and unadjusted R-Squared values, and Mallows' Cp statistic. To determine which models are best according to these different criteria, select one of the Tabular Options.



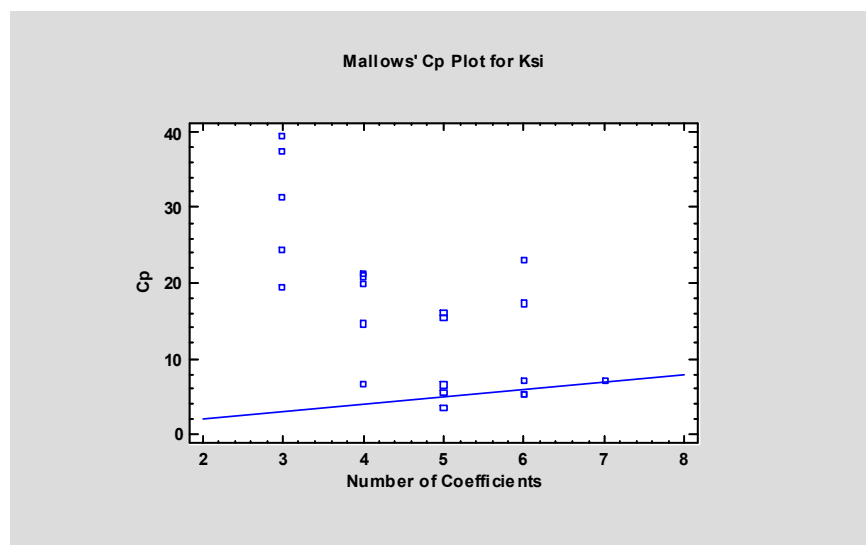
This plot shows the models which give the largest adjusted R-Squared values. The adjusted R-Squared statistic measures the proportion of the variability in Ksi which is explained by the model. Larger values of adjusted R-Squared correspond to smaller values of the mean squared error (MSE). Up to 5 models in each subset of between 2 and 6 variables are shown. The best model contains 4 variables, VWC, Soil_T, zrážky[mm], and Najnižšia vlhkosť[%]. To label each point or reduce the number of models displayed on the plot, press the alternate mouse button and select Pane Options.

Models with Largest Adjusted R-Squared

MSE	R-Squared	Adjusted R-Squared	Cp	Included Variables
0,000065393	98,7214	97,0165	3,39299	ABCF
0,0000878581	98,8547	95,9916	5,24769	ABCEF
0,0000896961	98,8308	95,9077	5,27379	ABCDF
0,000140833	99,0821	93,5747	7,0	ABCDEF
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0,000213148	97,2216	90,2754	7,02696	ABCDE
0,000233562	93,9109	89,3441	6,63373	ABC
0,00051599	86,5478	76,4587	14,6554	ACE
0,000605707	80,261	72,3655	19,5044	AC
0,000628593	87,7091	71,3213	15,3902	ACEF
0,000658711	87,1202	69,9472	16,0318	ACDE
0,000696133	81,8514	68,2399	19,7719	ACF
0,000732686	80,8984	66,5722	20,8101	BCE
0,00074102	75,8514	66,192	24,3085	CE
0,00074911	80,4702	65,8229	21,2765	ACD
0,000936865	69,4692	57,2569	31,2615	BC
0,000942875	87,7093	56,9827	17,39	ACDEF
0,00111333	63,7184	49,2057	37,5267	CF
0,00116625	61,9938	46,7913	39,4055	CD
0,0013403	82,5288	38,8507	23,0339	BCDEF

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This table shows the models which give the largest adjusted R-Squared values. The adjusted R-Squared statistic measures the proportion of the variability in Ksi which is explained by the model. Larger values of adjusted R-Squared correspond to smaller values of the mean squared error (MSE). Up to 5 models in each subset of between 2 and 6 variables are shown. The best model contains 4 variables, VWC, Soil_T, zrážky[mm], and Najnižšia vlhkosť[%].



This plot shows the models which give the smallest values of Mallows' Cp statistic. Cp is a measure of the bias in a model, based on a comparison of total mean squared error to the true error variance. Unbiased models have an expected value of approximately p, where p is the number of coefficients in the fitted model (including the constant). You should look for models with Cp values close to p. Unbiased models should be close to the line shown on the plot. To label each point or reduce the number of models displayed on the plot, press the alternate mouse button and select Pane Options.

Regression Model Selection - Ksi

Dependent variable: Ksi

Independent variables:

A=VWC

B=Soil_T

C=zrážky[mm]

D=Najvyššia teplota

E=Max. rýchlosť vetra[m/s]

F=Najnižšia vlhkosť[%]

Number of complete cases: 8

Number of models fit: 57

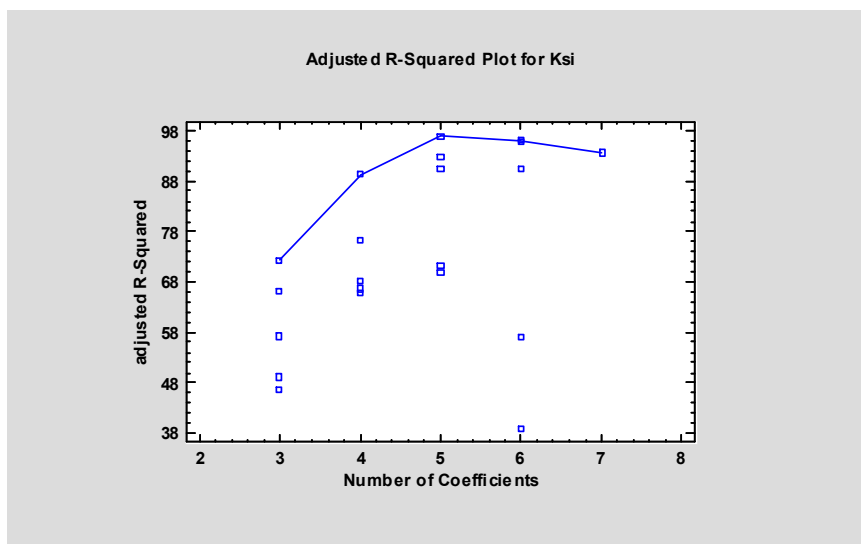
Model Results

<i>MSE</i>	<i>R-Squared</i>	<i>Adjusted R-Squared</i>	<i>Cp</i>	<i>Included Variables</i>
0,00213819	30,3199	2,44791	73,9123	AB
0,000605707	80,261	72,3655	19,5044	AC
0,00219185	28,5714	0,0	90,8841	AD
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0,000233562	93,9109	89,3441	6,63373	ABC
0,00219185	42,8571	0,0	73,8426	ABD
0,00181727	52,6226	17,0895	51,6149	ABE
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0,0000896961	98,8308	95,9077	5,27379	ABCDF
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This table shows the results of fitting various multiple regression models to describe the relationship between Ksi and 6 predictor variables. Models have been fit containing all combinations of from 2 to 6 variables. The statistics tabulated include the mean squared error (MSE), the adjusted and unadjusted R-Squared values, and Mallows' Cp statistic. To determine which models are best according to these different criteria, select one of the Tabular Options.



This plot shows the models which give the largest adjusted R-Squared values. The adjusted R-Squared statistic measures the proportion of the variability in Ksi which is explained by the model. Larger values of adjusted R-Squared correspond to smaller values of the mean squared error (MSE). Up to 5 models in each subset of between 2 and 6 variables are shown. The best model contains 4 variables, VWC, Soil_T, zrážky[mm], and Najnižšia vlhkosť[%]. To label each point or reduce the number of models displayed on the plot, press the alternate mouse button and select Pane Options.

Models with Largest Adjusted R-Squared

MSE	R-Squared	Adjusted R-Squared	Cp	Included Variables
0,000065393	98,7214	97,0165	3,39299	ABCF
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0,000732686	80,8984	66,5722	20,8101	BCE
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0,00116625	61,9938	46,7913	39,4055	CD
0,0013403	82,5288	38,8507	23,0339	BCDEF

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This table shows the models which give the largest adjusted R-Squared values. The adjusted R-Squared statistic measures the proportion of the variability in Ksi which is explained by the model. Larger values of adjusted R-Squared correspond to smaller values of the mean squared error (MSE). Up to 5 models in each subset of between 2 and 6 variables are shown. The best model contains 4 variables, VWC, Soil_T, zrážky[mm], and Najnižšia vlhkosť[%].

Models with Largest Adjusted R-Squared

MSE	R-Squared	Adjusted R-Squared	Cp	Included Variables
0,000065393	98,7214	97,0165	3,39299	ABCF
0,0000878581	98,8547	95,9916	5,24769	ABCEF
0,0000896961	98,8308	95,9077	5,27379	ABCDF
0,000140833	99,0821	93,5747	7,0	ABCDEF

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0,000605707	80,261	72,3655	19,5044	AC
0,000628593	87,7091	71,3213	15,3902	ACEF
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This table shows the models which give the largest adjusted R-Squared values. The adjusted R-Squared statistic measures the proportion of the variability in Ksi which is explained by the model. Larger values of adjusted R-Squared correspond to smaller values of the mean squared error (MSE). Up to 5 models in each subset of between 2 and 6 variables are shown. The best model contains 4 variables, VWC, Soil_T, zrážky[mm], and Najnižšia vlhkosť[%].

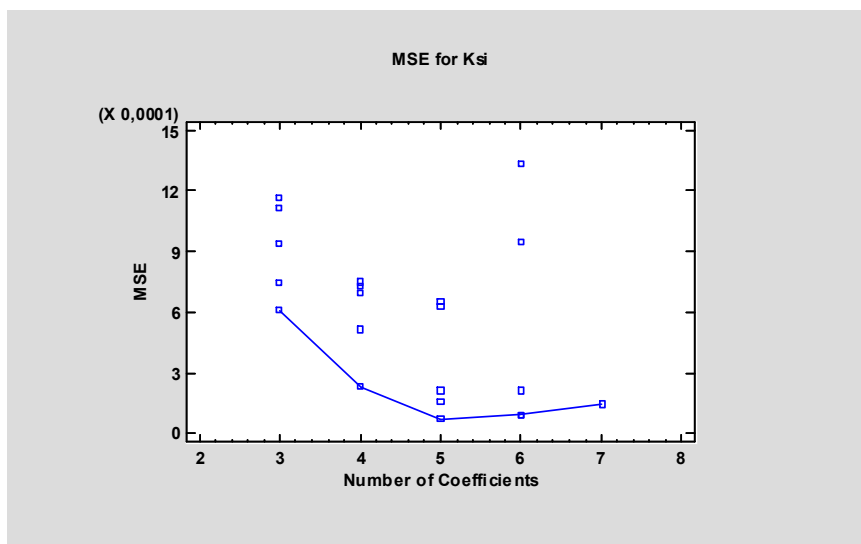
This plot shows the models which give the smallest values of Mallows' Cp statistic. Cp is a measure of the bias in a model, based on a comparison of total mean squared error to the true error variance. Unbiased models have an expected value of approximately p, where p is the number of coefficients in the fitted model (including the constant). You should look for models with Cp values close to p. Unbiased models should be close to the line shown on the plot. To label each point or reduce the number of models displayed on the plot, press the alternate mouse button and select Pane Options.

Models with Smallest Cp

<i>MSE</i>	<i>R-Squared</i>	<i>Adjusted R-Squared</i>	<i>Cp</i>	<i>Included Variables</i>
0,000065393	98,7214	97,0165	3,39299	ABCF
0,0000878581	98,8547	95,9916	5,24769	ABCEF
0,0000896961	98,8308	95,9077	5,27379	ABCDF
0,000157573	96,919	92,811	5,35658	ABCE
0,000212526	95,8445	90,3038	6,52719	ABCD
0,000233562	93,9109	89,3441	6,63373	ABC
0,000140833	99,0821	93,5747	7,0	ABCDEF
0,000213148	97,2216	90,2754	7,02696	ABCDE
0,00051599	86,5478	76,4587	14,6554	ACE
0,000628593	87,7091	71,3213	15,3902	ACEF
0,000658711	87,1202	69,9472	16,0318	ACDE
0,000942875	87,7093	56,9827	17,39	ACDEF
0,000605707	80,261	72,3655	19,5044	AC
0,000696133	81,8514	68,2399	19,7719	ACF
0,000732686	80,8984	66,5722	20,8101	BCE
0,00074911	80,4702	65,8229	21,2765	ACD
0,0013403	82,5288	38,8507	23,0339	BCDEF
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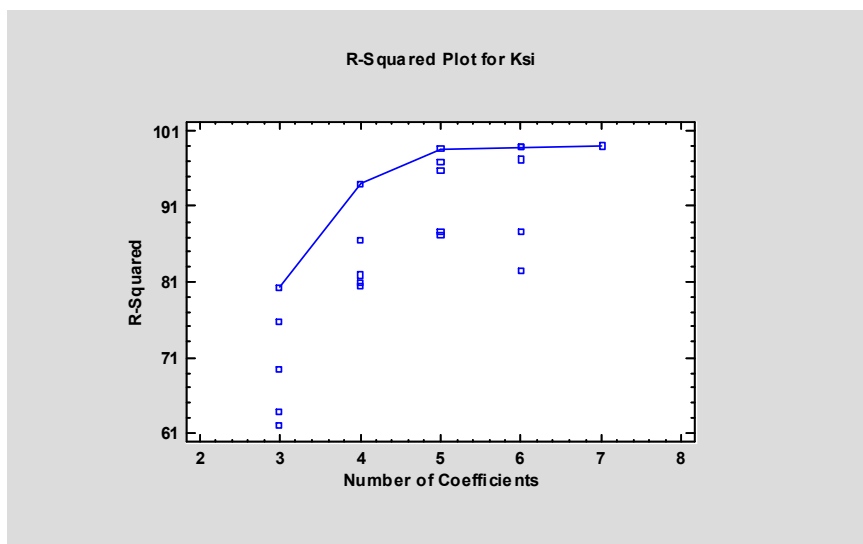
This plot shows the models which minimize the mean squared error. The mean squared error estimates the variance of the residuals. Better models typically give smaller values of MSE. Up to 5 models in each subset of between 2 and 6 variables are shown. The best model contains 4 variables, VWC, Soil_T, zrážky[mm], and Najnižšia vlhkosť[%]. To label each point or reduce the number of models displayed on the plot, press the alternate mouse button and select Pane Options.

Models with Best Information Criteria

MSE	Coefficients	AIC	HQC	SBIC	Included Variables
0,000065393	5	-8,38509	-8,71997	-8,33544	ABCF
0,0000878581	6	-7,83979	-8,24164	-7,78021	ABCEF
0,0000896961	6	-7,81908	-8,22093	-7,7595	ABCDF
0,000157573	5	-7,50562	-7,8405	-7,45597	ABCE
0,000233562	4	-7,36206	-7,62996	-7,32234	ABC
0,000212526	5	-7,20645	-7,54132	-7,1568	ABCD
0,000140833	7	-7,11794	-7,58676	-7,04842	ABCDEF
0,000213148	6	-6,95352	-7,35538	-6,89394	ABCDE
0,000605707	3	-6,65911	-6,86004	-6,62932	AC
0,00051599	4	-6,56942	-6,83732	-6,5297	ACE
0,00074102	3	-6,45748	-6,65841	-6,42769	CE
0,000696133	4	-6,26997	-6,53787	-6,23025	ACF
0,000936865	3	-6,22297	-6,4239	-6,19318	BC
0,000732686	4	-6,21879	-6,48669	-6,17907	BCE
0,00074911	4	-6,19662	-6,46453	-6,1569	ACD
0,000628593	5	-6,12203	-6,4569	-6,07238	ACEF
0,000658711	5	-6,07522	-6,4101	-6,02557	ACDE
0,00111333	3	-6,0504	-6,25132	-6,02061	CF
0,00116625	3	-6,00396	-6,20488	-5,97417	CD
0,000942875	6	-5,46658	-5,86843	-5,407	ACDEF
0,0013403	6	-5,11486	-5,51671	-5,05528	BCDEF

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This table sorts the regression models according to the value of the Akaike Information Criterion (AIC). The information criterion is based on the residual mean squared error with a penalty that grows as the number of model coefficients increases. The goal is to select a model with small residual error and as few coefficients as possible. The best model is the one that minimizes the information criterion. Often, the best model depends on the information criteria selected, which each use a different formula for the penalty.



This plot shows the models which give the largest values of R-Squared. R-Squared measures the proportion of the variability in Ksi which is explained by the model. Unfortunately, R-Squared always increases as additional variables are added to the model. To pick a good model, it is better to use the adjusted R-Squared statistic. To label each point or reduce the number of models displayed on the plot, press the alternate mouse button and select Pane Options.

Multiple Regression - Ksi

Dependent variable: Ksi

Independent variables:

VWC

Soil_T

zrážky[mm]

Najvyššia teplota

Max. rýchlosť vetra[m/s]

Najnižšia vlhkosť[%]

Number of observations: 8

		Standard	T	
Parameter	Estimate	Error	Statistic	P-Value
CONSTANT	-0,0765402	0,142537	-0,536984	0,6863
VWC	-0,00105666	0,000248823	-4,24663	0,1472
Soil_T	0,00853569	0,00242496	3,51994	0,1762
zrážky[mm]	-0,0423376	0,00638838	-6,62728	0,0953
Najvyššia teplota	0,00167299	0,00336154	0,497686	0,7060
Max. rýchlosť vetra[m/s]	0,00391758	0,00748699	0,523252	0,6931
Najnižšia vlhkosť[%]	0,00117669	0,000826495	1,42371	0,3898

Analysis of Variance

Source	Sum of Squares	Df	Mean Square	F-Ratio	P-Value
Model	0,0152021	6	0,00253368	17,99	0,1761
Residual	0,000140833	1	0,000140833		
Total (Corr.)	0,0153429	7			

R-squared = 99,0821 percent

R-squared (adjusted for d.f.) = 93,5747 percent

Standard Error of Est. = 0,0118673

Mean absolute error = 0,00356148

Durbin-Watson statistic = 1,70188

Lag 1 residual autocorrelation = -0,126381

The StatAdvisor

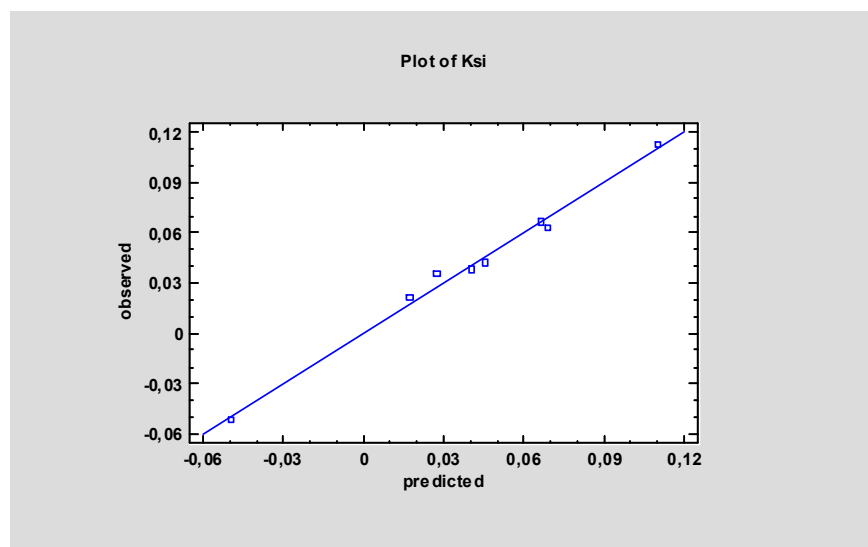
The output shows the results of fitting a multiple linear regression model to describe the relationship between Ksi and 6 independent variables. The equation of the fitted model is

Ksi = -0,0765402 - 0,00105666*VWC + 0,00853569*Soil_T - 0,0423376*zrážky[mm] + 0,00167299*Najvyššia teplota + 0,00391758*Max. rýchlosť vetra[m/s] + 0,00117669*Najnižšia vlhkosť[%]

Since the P-value in the ANOVA table is greater or equal to 0,05, there is not a statistically significant relationship between the variables at the 95,0% or higher confidence level.

The R-Squared statistic indicates that the model as fitted explains 99,0821% of the variability in Ksi. The adjusted R-squared statistic, which is more suitable for comparing models with different numbers of independent variables, is 93,5747%. The standard error of the estimate shows the standard deviation of the residuals to be 0,0118673. This value can be used to construct prediction limits for new observations by selecting the Reports option from the text menu. The mean absolute error (MAE) of 0,00356148 is the average value of the residuals. The Durbin-Watson (DW) statistic tests the residuals to determine if there is any significant correlation based on the order in which they occur in your data file.

In determining whether the model can be simplified, notice that the highest P-value on the independent variables is 0,7060, belonging to Najvyššia teplota. Since the P-value is greater or equal to 0,05, that term is not statistically significant at the 95,0% or higher confidence level. Consequently, you should consider removing Najvyššia teplota from the model.



Further ANOVA for Variables in the Order Fitted

Source	Sum of Squares	Df	Mean Square	F-Ratio	P-Value
VWC	0,00122907	1	0,00122907	8,73	0,2078
Soil_T	0,0034229	1	0,0034229	24,30	0,1274
zrážky[mm]	0,00975672	1	0,00975672	69,28	0,0761
Najvyššia teplota	0,000296671	1	0,000296671	2,11	0,3841
Max. rýchlosť vetra[m/s]	0,000211282	1	0,000211282	1,50	0,4359
Najnižšia vlhkosť[%]	0,000285463	1	0,000285463	2,03	0,3898
Model	0,0152021	6			

The StatAdvisor

This table shows the statistical significance of each variable as it was added to the model. You can use this table to help determine how much the model could be simplified, especially if you are fitting a polynomial.

95,0% confidence intervals for coefficient estimates

Parameter	Estimate	Standard		
		Error	Lower Limit	Upper Limit
CONSTANT	-0,0765402	0,142537	-1,88765	1,73457
VWC	-0,00105666	0,000248823	-0,00421826	0,00210494
Soil_T	0,00853569	0,00242496	-0,0222763	0,0393477
zrážky[mm]	-0,0423376	0,00638838	-0,12351	0,0388345
Najvyššia teplota	0,00167299	0,00336154	-0,0410394	0,0443854
Max. rýchlosť vet	0,00391758	0,00748699	-0,0912136	0,0990488
Najnižšia vlhkosť	0,00117669	0,000826495	-0,00932492	0,0116783

The StatAdvisor

This table shows 95,0% confidence intervals for the coefficients in the model. Confidence intervals show how precisely the

coefficients can be estimated given the amount of available data and the noise which is present.

Correlation matrix for coefficient estimates

	CONSTANT	VWC	Soil_T	zrážky[mm]
CONSTANT	1,0000	0,3320	-0,5903	-0,1967
VWC	0,3320	1,0000	-0,5268	-0,2347
Soil_T	-0,5903	-0,5268	1,0000	0,2733
zrážky[mm]	-0,1967	-0,2347	0,2733	1,0000
Najvyššia teplota	-0,7913	0,0304	0,1455	-0,0918
Max. rýchlosť vetra[m/s]	-0,8181	-0,5930	0,6612	0,4413
Najnižšia vlhkosť[%]	-0,9443	-0,4091	0,5885	0,0649

	Najvyššia teplota	Max. rýchlosť vetra[m/s]
CONSTANT	-0,7913	-0,8181
VWC	0,0304	-0,5930
Soil_T	0,1455	0,6612
zrážky[mm]	-0,0918	0,4413
Najvyššia teplota	1,0000	0,3278
Max. rýchlosť vetra[m/s]	0,3278	1,0000
Najnižšia vlhkosť[%]	0,6545	0,8376

	Najnižšia vlhkosť[%]
CONSTANT	-0,9443
VWC	-0,4091
Soil_T	0,5885
zrážky[mm]	0,0649
Najvyššia teplota	0,6545
Max. rýchlosť vetra[m/s]	0,8376
Najnižšia vlhkosť[%]	1,0000

The StatAdvisor

This table shows estimated correlations between the coefficients in the fitted model. These correlations can be used to detect the presence of serious multicollinearity, i.e., correlation amongst the predictor variables. In this case, there are 6 correlations with absolute values greater than 0,5 (not including the constant term).

Regression Results for Ksi

	Fitted	Std. Error	Lower 95,0%	Upper 95,0%	Lower 95,0%
Row	Value	CL for Forecast	CL for Forecast	CL for Forecast	CL for Mean

	Upper 95,0%
Row	CL for Mean

The StatAdvisor

This table contains information about Ksi generated using the fitted model. The table includes:

(1) the predicted value of Ksi using the fitted model (2) the standard error for each predicted value (3) 95,0% prediction limits for new observations (4) 95,0% confidence limits for the mean response

Each item corresponds to the values of the independent variables in a specific row of your data file. To generate forecasts for additional combinations of the variables, add additional rows to the bottom of your data file. In each new row, enter values for the independent variables but leave the cell for the dependent variable empty. When you return to this pane, forecasts will be added to the table for the new rows, but the model will be unaffected.

Cannot compute Studentized residuals.

Cannot compute influence measures.

Multiple Regression - Ksi

Dependent variable: Ksi

Independent variables:

VWC

Soil_T

zrážky[mm]

Najvyššia teplota

Max. rýchlosť vetra[m/s]
 Najnižšia vlhkosť[%]
 Number of observations: 8

		Standard	T	
Parameter	Estimate	Error	Statistic	P-Value
CONSTANT	0,0663857	0,0137157	4,84011	0,0029
zrážky[mm]	-0,0339092	0,0108602	-3,12233	0,0205

Analysis of Variance

Source	Sum of Squares	Df	Mean Square	F-Ratio	P-Value
Model	0,00949761	1	0,00949761	9,75	0,0205
Residual	0,00584532	6	0,00097422		
Total (Corr.)	0,0153429	7			

R-squared = 61,9022 percent
 R-squared (adjusted for d.f.) = 55,5526 percent
 Standard Error of Est. = 0,0312125
 Mean absolute error = 0,018367
 Durbin-Watson statistic = 3,0553 (P=0,9748)
 Lag 1 residual autocorrelation = -0,543541

Stepwise regression

Method: forward selection
 P-to-enter: 0,05
 P-to-remove: 0,05

Step 0:

0 variables in the model. 7 d.f. for error.
 R-squared = 0,00% Adjusted R-squared = 0,00% MSE = 0,00219185

Step 1:

Adding variable zrážky[mm] with P-to-enter = 0,020524
 1 variables in the model. 6 d.f. for error.
 R-squared = 61,90% Adjusted R-squared = 55,55% MSE = 0,00097422

Final model selected.

The StatAdvisor

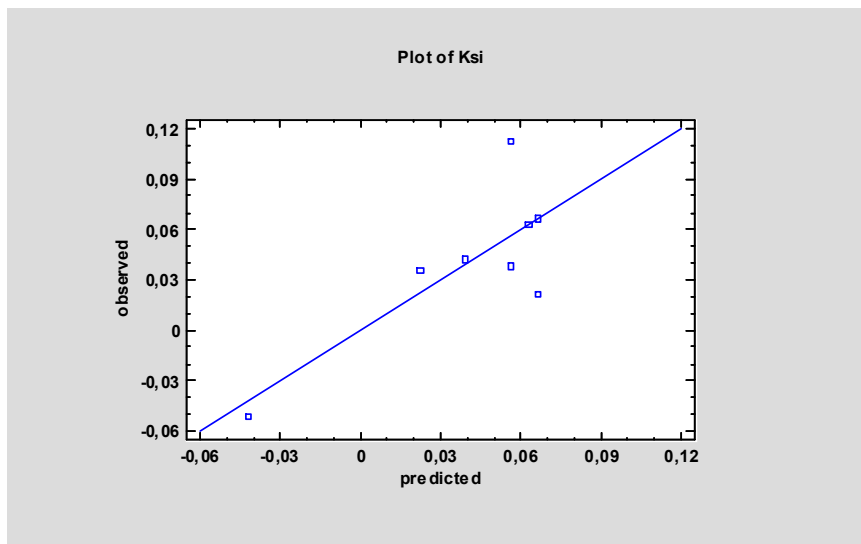
The output shows the results of fitting a multiple linear regression model to describe the relationship between Ksi and 6 independent variables. The equation of the fitted model is

$$Ksi = 0,0663857 - 0,0339092 * zrážky[mm]$$

Since the P-value in the ANOVA table is less than 0,05, there is a statistically significant relationship between the variables at the 95,0% confidence level.

The R-Squared statistic indicates that the model as fitted explains 61,9022% of the variability in Ksi. The adjusted R-squared statistic, which is more suitable for comparing models with different numbers of independent variables, is 55,5526%. The standard error of the estimate shows the standard deviation of the residuals to be 0,0312125. This value can be used to construct prediction limits for new observations by selecting the Reports option from the text menu. The mean absolute error (MAE) of 0,018367 is the average value of the residuals. The Durbin-Watson (DW) statistic tests the residuals to determine if there is any significant correlation based on the order in which they occur in your data file. Since the P-value is greater than 0,05, there is no indication of serial autocorrelation in the residuals at the 95,0% confidence level.

In determining whether the model can be simplified, notice that the highest P-value on the independent variables is 0,0205, belonging to zrážky[mm]. Since the P-value is less than 0,05, that term is statistically significant at the 95,0% confidence level. Consequently, you probably don't want to remove any variables from the model.



Further ANOVA for Variables in the Order Fitted

Source	Sum of Squares	Df	Mean Square	F-Ratio	P-Value
zrážky[mm]	0,00949761	1	0,00949761	9,75	0,0205
Model	0,00949761	1			

The StatAdvisor

This table shows the statistical significance of each variable as it was added to the model. You can use this table to help determine how much the model could be simplified, especially if you are fitting a polynomial.

95,0% confidence intervals for coefficient estimates

Parameter	Estimate	Standard Error	Lower Limit	Upper Limit
CONSTANT	0,0663857	0,0137157	0,0328244	0,099947
zrážky[mm]	-0,0339092	0,0108602	-0,0604832	-0,0073351

The StatAdvisor

This table shows 95,0% confidence intervals for the coefficients in the model. Confidence intervals show how precisely the coefficients can be estimated given the amount of available data and the noise which is present.

Correlation matrix for coefficient estimates

	CONSTANT	zrážky[mm]
CONSTANT	1,0000	-0,5939
zrážky[mm]	-0,5939	1,0000

The StatAdvisor

This table shows estimated correlations between the coefficients in the fitted model. These correlations can be used to detect the presence of serious multicollinearity, i.e., correlation amongst the predictor variables. In this case, there are no correlations with absolute values greater than 0,5 (not including the constant term).

Regression Results for Ksi

	Fitted Value	Std. Error	Lower 95,0% CL for Forecast	Upper 95,0% CL for Forecast	Lower 95,0% CL for Mean
Row					
	Upper 95,0% CL for Mean				
Row					

The StatAdvisor

This table contains information about Ksi generated using the fitted model. The table includes:

- (1) the predicted value of Ksi using the fitted model
- (2) the standard error for each predicted value
- (3) 95,0% prediction limits for new observations
- (4) 95,0% confidence limits for the mean response

Each item corresponds to the values of the independent variables in a specific row of your data file. To generate forecasts for additional combinations of the variables, add additional rows to the bottom of your data file. In each new row, enter values for the independent variables but leave the cell for the dependent variable empty. When you return to this pane, forecasts will be added to the table for the new rows, but the model will be unaffected.

Unusual Residuals

		<i>Predicted</i>		<i>Studentized</i>
Row	Y	Y	Residual	Residual
4	0,112542	0,0562129	0,0563291	2,97

The StatAdvisor

The table of unusual residuals lists all observations which have Studentized residuals greater than 2 in absolute value. Studentized residuals measure how many standard deviations each observed value of Ksi deviates from a model fitted using all of the data except that observation. In this case, there is one Studentized residual greater than 2, but none greater than 3.

Influential Points

		<i>Mahalanobis</i>	
Row	Leverage	Distance	DFITS
3	0,851695	33,6	-1,86843
4	0,149516	0,197661	1,24534

Average leverage of single data point = 0,25

The StatAdvisor

The table of influential data points lists all observations which have leverage values greater than 3 times that of an average data point, or which have an unusually large value of DFITS. Leverage is a statistic which measures how influential each observation is in determining the coefficients of the estimated model. DFITS is a statistic which measures how much the estimated coefficients would change if each observation was removed from the data set. In this case, an average data point would have a leverage value equal to 0,25. There is one data point with more than 3 times the average leverage, but none with more than 5 times. There are 2 data points with unusually large values of DFITS.

Multiple Regression - Ksi

Dependent variable: Ksi

Independent variables:

VWC

Soil_T

zrážky[mm]

Najvyššia teplota

Max. rýchlosť vetra[m/s]

Najnižšia vlhkosť[%]

Number of observations: 8

		<i>Standard</i>	<i>T</i>	
Parameter	Estimate	Error	Statistic	P-Value
CONSTANT	0,0117247	0,0150837	0,777307	0,4936
VWC	-0,000999895	0,000130435	-7,66585	0,0046
Soil_T	0,00775988	0,00123342	6,29138	0,0081
zrážky[mm]	-0,043261	0,0037513	-11,5323	0,0014
Najnižšia vlhkosť[%]	0,000699722	0,000208278	3,35956	0,0437

Analysis of Variance

Source	Sum of Squares	Df	Mean Square	F-Ratio	P-Value
Model	0,0151468	4	0,00378669	57,91	0,0036
Residual	0,000196179	3	0,000065393		
Total (Corr.)	0,0153429	7			

R-squared = 98,7214 percent

R-squared (adjusted for d.f.) = 97,0165 percent

Standard Error of Est. = 0,0080866

Mean absolute error = 0,00449011

Durbin-Watson statistic = 2,27809 (P=0,7207)

Lag 1 residual autocorrelation = -0,308391

Stepwise regression

Method: backward selection

P-to-enter: 0,05

P-to-remove: 0,05

Step 0:

6 variables in the model. 1 d.f. for error.

R-squared = 99,08% Adjusted R-squared = 93,57% MSE = 0,000140833

Step 1:

Removing variable Najvyššia teplota with P7-to-remove =0,706012

5 variables in the model. 2 d.f. for error.

R-squared = 98,85% Adjusted R-squared = 95,99% MSE = 0,0000878581

Step 2:

Removing variable Max. rýchlosť vetra[m/s] with P7-to-remove =0,677033

4 variables in the model. 3 d.f. for error.

R-squared = 98,72% Adjusted R-squared = 97,02% MSE = 0,000065393

Final model selected.

The StatAdvisor

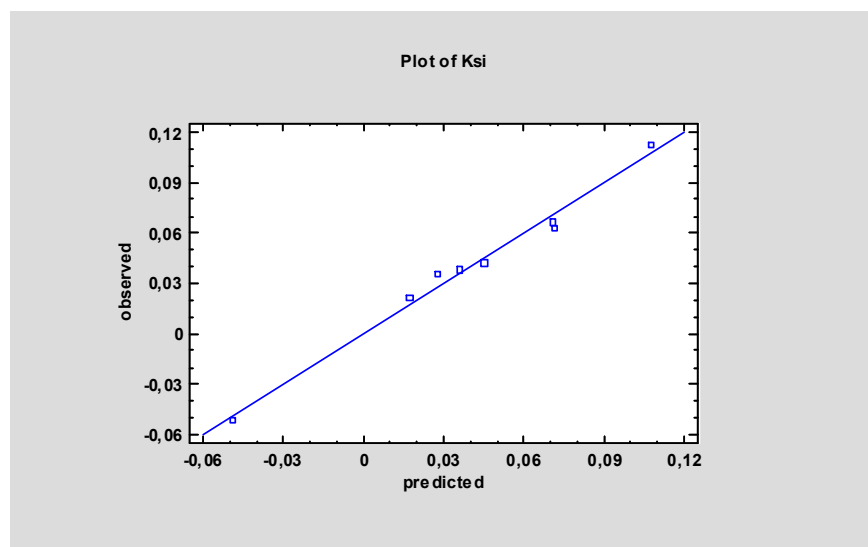
The output shows the results of fitting a multiple linear regression model to describe the relationship between Ksi and 6 independent variables. The equation of the fitted model is

$$\text{Ksi} = 0,0117247 - 0,000999895 \cdot \text{VWC} + 0,00775988 \cdot \text{Soil_T} - 0,043261 \cdot \text{zrážky[mm]} + 0,000699722 \cdot \text{Najnižšia vlhkos?[\%]}$$

Since the P-value in the ANOVA table is less than 0,05, there is a statistically significant relationship between the variables at the 95,0% confidence level.

The R-Squared statistic indicates that the model as fitted explains 98,7214% of the variability in Ksi. The adjusted R-squared statistic, which is more suitable for comparing models with different numbers of independent variables, is 97,0165%. The standard error of the estimate shows the standard deviation of the residuals to be 0,0080866. This value can be used to construct prediction limits for new observations by selecting the Reports option from the text menu. The mean absolute error (MAE) of 0,00449011 is the average value of the residuals. The Durbin-Watson (DW) statistic tests the residuals to determine if there is any significant correlation based on the order in which they occur in your data file. Since the P-value is greater than 0,05, there is no indication of serial autocorrelation in the residuals at the 95,0% confidence level.

In determining whether the model can be simplified, notice that the highest P-value on the independent variables is 0,0437, belonging to Najnižšia vlhkos?[%]. Since the P-value is less than 0,05, that term is statistically significant at the 95,0% confidence level. Consequently, you probably don't want to remove any variables from the model.



Further ANOVA for Variables in the Order Fitted

Source	Sum of Squares	Df	Mean Square	F-Ratio	P-Value
VWC	0,00122907	1	0,00122907	18,80	0,0226

Soil_T	0,0034229	1	0,0034229	52,34	0,0054
zrážky[mm]	0,00975672	1	0,00975672	149,20	0,0012
Najnižšia vlhkosť[%]	0,00073807	1	0,00073807	11,29	0,0437
Model	0,0151468	4			

The StatAdvisor

This table shows the statistical significance of each variable as it was added to the model. You can use this table to help determine how much the model could be simplified, especially if you are fitting a polynomial.

95,0% confidence intervals for coefficient estimates

	<i>Estimate</i>	<i>Standard Error</i>	<i>Lower Limit</i>	<i>Upper Limit</i>
CONSTANT	0,0117247	0,0150837	-0,0362785	0,0597279
VWC	-0,000999895	0,000130435	-0,001415	-0,000584792
Soil_T	0,00775988	0,00123342	0,0038346	0,0116852
zrážky[mm]	-0,043261	0,0037513	-0,0551993	-0,0313227
Najnižšia vlhkosť	0,000699722	0,000208278	0,000036889	0,00136255

The StatAdvisor

This table shows 95,0% confidence intervals for the coefficients in the model. Confidence intervals show how precisely the coefficients can be estimated given the amount of available data and the noise which is present.

Correlation matrix for coefficient estimates

	CONSTANT	VWC	Soil_T	zrážky[mm]
CONSTANT	1,0000	-0,1794	-0,7861	0,1930
VWC	-0,1794	1,0000	-0,2034	0,1305
Soil_T	-0,7861	-0,2034	1,0000	-0,0581
zrážky[mm]	0,1930	0,1305	-0,0581	1,0000
Najnižšia vlhkosť[%]	-0,6336	-0,0286	0,2356	-0,6405

	Najnižšia vlhkosť[%]
CONSTANT	-0,6336
VWC	-0,0286
Soil_T	0,2356
zrážky[mm]	-0,6405
Najnižšia vlhkosť[%]	1,0000

The StatAdvisor

This table shows estimated correlations between the coefficients in the fitted model. These correlations can be used to detect the presence of serious multicollinearity, i.e., correlation amongst the predictor variables. In this case, there is 1 correlation with absolute value greater than 0,5 (not including the constant term).

Regression Results for Ksi

	<i>Fitted Value</i>	<i>Std. Error</i>	<i>Lower 95,0%</i>	<i>Upper 95,0%</i>	<i>Lower 95,0%</i>
Row	Value	CL for Forecast	CL for Forecast	CL for Forecast	CL for Mean

	<i>Upper 95,0%</i>
Row	CL for Mean

The StatAdvisor

This table contains information about Ksi generated using the fitted model. The table includes:

(1) the predicted value of Ksi using the fitted model (2) the standard error for each predicted value (3) 95,0% prediction limits for new observations (4) 95,0% confidence limits for the mean response

Each item corresponds to the values of the independent variables in a specific row of your data file. To generate forecasts for additional combinations of the variables, add additional rows to the bottom of your data file. In each new row, enter values for the independent variables but leave the cell for the dependent variable empty. When you return to this pane, forecasts will be added to the table for the new rows, but the model will be unaffected.

Unusual Residuals

		<i>Predicted</i>		<i>Studentized</i>
Row	Y	Y	<i>Residual</i>	<i>Residual</i>

3	-0,0518175	-0,0492176	-0,00259992	-2,05
4	0,112542	0,107445	0,00509696	2,34
5	0,0211082	0,0173491	0,00375911	2,09

The StatAdvisor

The table of unusual residuals lists all observations which have Studentized residuals greater than 2 in absolute value. Studentized residuals measure how many standard deviations each observed value of Ksi deviates from a model fitted using all of the data except that observation. In this case, there are 3 Studentized residuals greater than 2, but none greater than 3.

Influential Points

		<i>Mahalanobis</i>	
Row	Leverage	Distance	DFITS
3	0,949154	111,145	-8,85918
4	0,819211	26,3306	4,98137
5	0,894976	50,2728	6,0999
8	0,911966	61,2984	-5,79819

Average leverage of single data point = 0,625

The StatAdvisor

The table of influential data points lists all observations which have leverage values greater than 3 times that of an average data point, or which have an unusually large value of DFITS. Leverage is a statistic which measures how influential each observation is in determining the coefficients of the estimated model. DFITS is a statistic which measures how much the estimated coefficients would change if each observation was removed from the data set. In this case, an average data point would have a leverage value equal to 0,625. There are no data points with more than 3 times the average leverage. There are 4 data points with unusually large values of DFITS.